This report presents the methodology and implementation of an automated pipeline for analyzing semi-structured text from Electronic Medical Records (EMRs) to support personalized clinical decision-making. The primary contribution is a two-stage approach combining topic modeling to identify thematic categories of medical recommendations, and intra-topic clustering to discover variations in physician practices and treatment approaches. The system leverages natural language processing (NLP) techniques including sentence segmentation, lemmatization, TF-IDF vectorization, Latent Dirichlet Allocation (LDA), and DBSCAN clustering. Experimental validation is performed on real-world clinical data from the Almazov National Medical Research Centre in St. Petersburg, Russia.

1. Introduction

1.1 Problem Statement

Electronic Medical Records contain vast amounts of unstructured and semi-structured textual data, including physician notes, diagnostic reports, and treatment recommendations. While this information is crucial for clinical decision support, its effective utilization faces several challenges:

- **Heterogeneity of data formats:** Records vary from single sentences to multi-paragraph documents
- **Semi-structured nature:** Clinical notes contain specialized terminology, abbreviations, and inconsistent formatting
- **Variability in physician practices:** Different doctors may provide similar recommendations using different language or approaches
- **Lack of personalization:** Existing clinical decision support systems often fail to account for individual physician preferences and practice patterns

1.2 Research Objectives

The primary objectives of this research are:

1. Develop a preprocessing pipeline for Russian-language clinical text that handles sentence segmentation, cleaning, and lemmatization
2. Implement topic modeling to identify thematic categories within physician recommendations
3. Create an intra-topic clustering methodology to identify distinct physician approaches and recommendation patterns
4. Validate the approach on real-world clinical data from a major medical center

1.3 Contributions

This work makes the following contributions:

- A comparative analysis of sentence segmentation tools for Russian medical text
- An end-to-end pipeline for processing semi-structured EMR data
- A novel two-stage analysis framework combining topic modeling with DBSCAN clustering
- Empirical evaluation on authentic clinical data from the Almazov Center

2. Methodology

2.1 Data Description

The dataset consists of clinical records from the Almazov National Medical Research Centre, specifically focusing on the "general recommendations" field from patient consultations. Each record contains:

Field	**Description**
Регистрационный номер	Registration number (patient identifier)
Номер эпизода	Episode number
ФИО специалиста	Physician's full name
Дата время выполнения	Date and time of consultation
рекомендации общие	General recommendations (free text)

The data exhibits significant variability in length, structure, and content complexity.

2.2 Preprocessing Pipeline

The preprocessing stage consists of several sequential steps, implemented in ResearchWorkPreprocessingAlmazovRazdel:

Sentence Segmentation

Four segmentation approaches were evaluated:

- **Razdel** (Natasha project) - Russian-specific rule-based tokenizer
- **spaCy** (ru_core_news_sm) - Statistical model for Russian
- **NLTK Punkt** - Unsupervised sentence boundary detection
- **Custom medical regex** - Rule-based approach with medical abbreviation handling

Based on experimental results **Razdel** was selected for the production pipeline due to its superior handling of medical abbreviations and multi-line records.

Text Cleaning and Lemmatization

Each segmented sentence undergoes:

1. **Lowercasing:** Convert all text to lowercase
2. **Special character removal:** Remove punctuation, numbers, and non-Cyrillic characters (except spaces)
3. **Stop word removal:** Filter common Russian stop words (nltk.corpus.stopwords)
4. **Lemmatization:** Convert words to their base form using spaCy's Russian model

The cleaned text is filtered to retain only segments with more than 2 words after processing.

Data Explosion

Since one clinical record may contain multiple recommendations, the processed segments are exploded into individual rows, preserving metadata (physician ID, timestamp, etc.) for downstream analysis.

2.3 Topic Modeling

Topic modeling is performed using Latent Dirichlet Allocation (LDA) with the following configuration:

```python
lda_model = LdaModel(
    corpus=corpus,
    id2word=dictionary,
    num_topics=11,  # Optimal value determined via coherence analysis
    random_state=42,
    passes=10,
    iterations=100,
    alpha='auto'
)
```

Optimal Number of Topics Selection

The optimal number of topics (k) was determined by:

- Evaluating coherence scores (c_v metric) for $k \in [2, 20]$
- Selecting $k = 11$ as the point where coherence score plateaued
- Visual validation using pyLDAvis to ensure topic interpretability

Topic Assignment

Each processed text segment is assigned a dominant topic based on the highest probability distribution from the LDA model.

2.4 Intra-Topic Clustering

After topic assignment, records within each topic are clustered using DBSCAN (Density-Based Spatial Clustering of Applications with Noise) to identify subgroups representing different physician approaches.

Feature Representation

Text within each topic is vectorized using TF-IDF (Term Frequency-Inverse Document Frequency):

```
python
vectorizer = TfidfVectorizer(max_df=0.95, min_df=2, max_features=1000)
tfidf_matrix = vectorizer.fit_transform(topic_texts)
```

DBSCAN Parameter Selection

The eps parameter (maximum distance between two samples to be considered neighbors) is determined dynamically using the **elbow method** on k-distance graphs (k=5). This approach adapts to the local density characteristics of each topic's feature space.

Quality Assessment

Cluster quality is evaluated using:

- **Silhouette score** (for clusters with >1 point, excluding noise)
- **Cluster diameter** (maximum intra-cluster distance in reduced space)
- **Visual inspection** using t-SNE and UMAP dimensionality reduction

2.5 Dimensionality Reduction for Visualization

Two techniques are employed for visualizing high-dimensional cluster structures:

Method	Characteristics	Use Case
t-SNE	Preserves local neighborhood structure	Detailed cluster separation analysis
UMAP	Preserves both local and global structure, faster	Large-scale pattern discovery

Both methods reduce the TF-IDF matrix to 3 dimensions for 3D scatter plot visualization using Plotly.

3. Experimental Results

3.1 Sentence Segmentation Comparison

Key finding: Razdel demonstrates superior performance, particularly in handling medical abbreviations and multi-line clinical notes (Figure 1).

```
================================================================================
COMPARISON OF SEGMENTATION TOOLS FOR MEDICAL DATA
================================================================================
```

Algorithm	F1-score	Precision	Recall	TP (matched)	FP (extra)	FN (missed)
Razdel	0.8350	0.7605	0.9257	635	200	51
spaCy (RU)	0.7955	0.6690	0.9810	673	333	13
Baseline (Medical)	0.7694	0.7347	0.8076	554	200	132
NLTK (Punkt)	0.7099	0.5757	0.9257	635	468	51
Custom Medical	0.5077	0.3741	0.7901	542	907	144

Figure 1 – Methods Comparing Table

3.2 Topic Modeling Results

The LDA model identified 11 interpretable topics, including but not limited to:

Topic matching:

Тема 0 (LDA, начиная с 0): ['ад', 'день', 'пульс', 'раз', 'контроль', 'дневник', 'мм', 'рт', 'ст', 'завтрак']

Topic 1 (LDA, starting from 0): ['неделя', 'умеренный', 'нагрузка', 'физический', 'динамический', 'аэробный', 'менее', 'день', 'регулярный', 'минута']

Topic 2 (LDA, starting from 0): ['самоконтроль', 'давление', 'артериальный', 'электронный', 'вес', 'ведение', 'снижение', 'условие', 'дневник', 'домашний']

Topic 3 (LDA, starting from 0): ['физический', 'упражнение', 'дозированные', 'мина', 'нагрузка', 'ежедневно', 'аэробный', 'активность', 'регулярный', 'ходьба']

Topic 4 (LDA, starting from 0): ['отдых', 'труд', 'режим', 'нормализация', 'тело', 'масса', 'кг', 'менее', 'индекс', 'курение']

Topic 5 (LDA, starting from 0): ['темп', 'день', 'ходьба', 'врач', 'ежедневный', 'визит', 'размеренный', 'принести', 'мина', 'тренировка']

Topic 6 (LDA, starting from 0): ['снижение', 'стрессовый', 'нагрузка', 'мг', 'ситуация', 'аналог', 'либо', 'приём', 'моксонидин', 'вода']

Topic 7 (LDA, starting from 0): ['целевой', 'кардиолог', 'интенсивный', 'жизнь', 'образ', 'сайт', 'подробный', 'общество', 'европейский', 'ознакомиться']

Topic 8 (LDA, starting from 0): ['ад', 'контроль', 'дневник', 'утро', 'пульс', 'вечер', 'чсс', 'измерение', 'запись', 'повторный']

Topic 9 (LDA, starting from 0): ['избегать', 'переутомление', 'эмоциональный', 'нагрузка', 'фн', 'переохлаждение', 'менее', 'пульс', 'эффективный', 'увеличение']

Topic 10 (LDA, starting from 0): ['нагрузка', 'физ', 'прогулка', 'ограничение', 'режим', 'физический', 'контроль', 'общий', 'рекомендация', 'тяжёлый']

Topic distribution:

dominant_topic	
10	11704
8	9542
4	6706
3	5845
6	5488
0	5309
1	4861
7	4646
9	4403
2	3237
5	2006